

AIVOA.AI

UI / UX Design & Screen Specification

Connected QMS interface and screen specification

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Professional presentation edition • Content preserved from the supplied source documents

12. Screen 08 — CAPA

[illegible]

13. Screen 09 — Effectiveness & Closure

CAPA EFFECTIVENESS CAPA-2026-009 Method Monitor next 5 API batches Success Criteria No recurrence of temperature excursion
Batch 042 ✓ No recurrence Batch 043 ✓ No recurrence Batch 044 ✓ No recurrence Batch 045 ✓ No recurrence Batch 046 ✓ No recurrence
RESULT ✓ EFFECTIVE Reviewer: QA Manager Reviewed: 29 Sep 2026 [Close CAPA]

After closure, the UI should show a read-only closure summary with who closed it, when, why and the effectiveness result. Normal edit controls disappear.

14. Screen 10 — Batch Release

BATCH RELEASE REVIEW API-2026-041 Manufacturing record ✓ Complete In-process checks ✓ Reviewed Deviation DEV-2026-018 ✓ Investigated CAPA CAPA-2026-009 ✓ Effectiveness reviewed QC results ✓ Available QUALITY REVIEW Deviation impact: Reviewed CAPA status: Effective Release recommendation: Pending QA decision [Release] [Quarantine] [Reject] Decision rationale _____ [Confirm Decision]

The release screen should demonstrate that the quality record informs the batch disposition. The final decision is a human quality action, not an AI action.

15. Screen 11 — Complaint / Recall Context

COM-2026-003 Product Quality Complaint Product: Finished Product Related API Batch: API-2026-041 Related Deviation: DEV-2026-018
Complaint Customer reports product quality issue. INVESTIGATION Status: Under Investigation RISK ASSESSMENT Potential product
impact: Review required DOWNSTREAM ACTION Recall assessment: Pending [View Batch] [View Deviation] [Open Recall Assessment
]

16. Screen 12 — Supplier Quality

SUPPLIER ChemCorp Status: Approved Risk: Medium RAW MATERIAL Acetic Anhydride Lot: RM-2026-001 BATCHES USING MATERIAL API-2026-041 QUALITY EVENTS 2 Open supplier event: 1 RELATED DEVIATION DEV-2026-018 [Open Supplier Quality Event]

17. The Most Important UI Component — Linked Records

LINKED RECORDS ██████████ Batch API-2026-041 → In-Process Check
IPC-2026-041-03 → Deviation DEV-2026-018 → Investigation INV-2026-012 → Root Cause RCA-2026-012 → CAPA CAPA-2026-009 →
Effectiveness EFF-2026-009 → Batch Release BR-2026-041 → ██████████

Put this card on deviation, investigation and CAPA detail pages. It is the visual proof that the product is a connected QMS rather than separate CRUD screens.

18. Universal Workflow Stepper

Reported → Investigation → Root Cause
CAPA
Effectiveness
Closed

19. Interaction and Modal Patterns

20. Notifications / Work Queue

ACTION REQUIRED

■ CAPA-2026-009 Preventive action due tomorrow ■ INV-2026-012 2 investigation tasks pending ■ API-2026-041 Batch release review pending ✓ DEV-2026-017 Closed successfully

Notifications should be actionable and tied to actual records. Do not create a generic notification center full of fake activity.

21. Responsive Behavior

The competition demo is likely desktop-first. Do not spend scarce time building a perfect mobile application unless the existing Round 1 product already requires it.

22. Accessibility and Usability

- Every status badge includes text, not color alone.
- All inputs have visible labels; placeholders are not the only labels.
- Keyboard focus states remain visible.
- Tables have clear column headings.
- Primary action is visually distinct from secondary actions.
- Destructive/closing actions use confirmation and clear consequences.
- AI-generated content is explicitly labeled as AI-generated until accepted.
- Evidence links should open source metadata without losing the current record context.

23. UI Design Image Set to Create

For the final design/presentation, create separate high-fidelity visuals rather than one giant collage. Each image should represent one screen and use the same component library.

23.1 Image-generation prompt template

High-fidelity enterprise SaaS UI for a pharmaceutical quality management system named PharmaOne-AI. Preserve a clean white background, dark navy sidebar, blue primary actions, subtle gray borders, rounded 12px cards, compact typography, professional regulated-industry feel. Show realistic pharmaceutical quality data, record IDs, status badges, tables and workflow steps. Do not use marketing-style illustrations. The screen must look like an actual working B2B application, with strong hierarchy and readable labels.

24. Prototype Navigation / Click Flow

Dashboard ↓ click batch Batch API-2026-041 ↓ click out-of-spec check In-Process Check ↓ Create Deviation Deviation DEV-2026-018 ↓ Start Investigation Investigation INV-2026-012 ↓ Root Cause tab 5 Whys / Root Cause ↓ Create CAPA CAPA-2026-009 ↓ Effectiveness Effectiveness Check ↓ Close CAPA Closed ↓ Batch link Batch Release Review ↓ Complaint link Complaint / Recall Context ■ Supplier link Supplier Quality

25. What to Build First in the UI

26. UI Acceptance Checklist

27. Final UX Principle

The evaluator should never have to guess what happens next. Every important quality screen should answer three questions: “What happened?”, “What is the current quality state?”, and “What is the next controlled action?” The visual workflow stepper, linked records, AI evidence panel and explicit human actions are the four components that make the Round 1 product feel like the connected QMS described by the Round 2 challenge.

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Design goal	Decision
Preserve Round 1	Keep current visual language, dashboard patterns, deviation form and AI assistant concept.
Round 2 focus	Make Deviation → Investigation → Root Cause → CAPA → Effectiveness → Closure feel like one continuous product.
Context	Add lightweight Batch, In-Process Checks, Batch Release, Complaint and Supplier views.
AI principle	AI recommends/extracts; human reviews and owns final quality decisions.
Demo principle	One connected story: API-2026-041 → DEV-2026-018 → INV-2026-012 → CAPA-2026-009 → closure.

Existing Round 1 element	Round 2 treatment
White/light application background	Keep
Dark/navy navigation	Keep
Blue primary action	Keep

Existing Round 1 element	Round 2 treatment
Compact enterprise cards	Keep
Deviation severity badges	Keep and extend to lifecycle status
AI Assistant panel	Rename/extend to AI Quality Assistant
Deviation form	Keep; add batch/process context
Dashboard metrics	Expand from deviation-only to quality lifecycle metrics

Global component	Behavior
Sidebar	Persistent, collapsible on smaller screens; active module clearly highlighted
Top search	Search record references, batch numbers, deviations and CAPAs
Alerts	Overdue investigation/CAPA tasks and pending reviews
User menu	Profile, role, sign out
Breadcrumb	Dashboard / Deviations / DEV-2026-018 / Investigation
Page actions	Primary action on right; destructive/closing actions visually separated
Record reference	Always show human-readable ID prominently

Token	Recommendation	Use
Background	#F7F9FC / white	App and cards
Primary	Deep navy/blue	Navigation and primary actions
Success	Green	Completed, effective, approved
Warning	Amber	Pending, at risk, due soon
Danger	Red	Major/critical, overdue, out-of-spec
Neutral	Slate/gray	Draft, disabled, metadata
Radius	10–14 px	Cards and inputs
Border	1 px light slate	Card separation
Typography	Inter/Aptos-like sans	Readable enterprise UI
Spacing	8 px base grid	Consistent vertical rhythm

State	UI appearance	Primary action
Reported	Blue/current or completed	Start Investigation
Investigation	Blue/current	Complete Investigation
Root Cause	Blue/current	Approve Root Cause
CAPA	Blue/current	Complete Actions
Effectiveness	Blue/current	Record Result
Closed	Green/completed	Read-only

Interaction	UI pattern	Why
Create deviation from check	Prefill batch/check/parameter	Avoid duplicate data entry
AI recommendation	Side panel + Accept/Edit/Reject	Human control
Start investigation	Confirmation modal showing linked deviation	Prevent accidental workflow start
Complete investigation	Validation summary modal	Catch missing evidence/conclusion
Create CAPA	Pre-populate root cause + source links	Maintain traceability
Close CAPA	Effectiveness + closure checklist modal	Prevent premature closure
Batch release	Decision modal + rationale + e-sign placeholder	Make decision explicit
Delete	Avoid normal delete; archive/deactivate if needed	Protect quality history

Breakpoint	Behavior
Desktop ≥ 1280px	Persistent sidebar + two-column detail pages + AI side panel
Tablet 768–1279px	Collapsible sidebar; detail pages become stacked sections
Mobile < 768px	Single-column record view; AI assistant becomes bottom sheet/drawer; tables become cards

Image	Screen	What must be visible
01	QMS Dashboard	Metrics, trend, recent events, actions required
02	Batch Detail	Raw material, manufacturing steps, in-process checks, linked deviation
03	Deviation Detail	Lifecycle stepper, event data, AI assistant, evidence, linked records
04	Investigation	Tasks, evidence, AI investigation assistance, tabs
05	Root Cause	5 Whys, AI draft, reviewer controls, final root cause
06	CAPA	Corrective/preventive actions, owners, dates, AI suggestions
07	Effectiveness	Success criteria, monitored batches, result, closure action
08	Batch Release	Quality review checklist and release/quarantine/reject decision
09	Complaint	Complaint, linked batch/deviation, risk and recall assessment
10	Supplier	Supplier, material lot, quality events and linked deviation
11	Connected workflow	End-to-end diagram from batch → deviation → investigation → CAPA → closure

Priority	Screen/component	Depth
P0	Workflow Stepper	Deep/reusable
P0	Deviation Detail upgrade	Deep
P0	Investigation Workspace	Deep
P0	Root Cause / 5 Whys	Deep
P0	CAPA	Deep
P0	Effectiveness / Closure	Deep
P1	Dashboard upgrade	Medium
P1	Batch + In-Process Checks	Medium
P1	Batch Release	Light
P1	Complaint	Light
P1	Supplier	Light
P2	Adverse Event / Recall	Very light
P2	Advanced reports	Skip if time is limited

Requirement	UI passes when...
Manufacturing context	User can open a batch and see raw material, manufacturing step and in-process checks
Deviation trigger	Out-of-spec check can create a prefilled deviation
AI assistance	AI extraction/recommendation is visible with evidence and human review controls
Lifecycle	Deviation shows Reported → Investigation → Root Cause → CAPA → Effectiveness → Closed
Investigation	Tasks and evidence are visible and actionable
Root cause	5 Whys can be reviewed and final root cause explicitly saved
CAPA	Corrective and preventive actions have owner/due/status
Effectiveness	Success criteria and results are visible before closure
Closure	Closed record is read-only with actor/time/result
Batch release	Reviewer sees linked quality state before decision
Connections	Complaint and supplier records link back to batch/deviation
Consistency	All screens share the same visual language and navigation